

A look at the wireless charging product range

- Mobile phones capable of wireless charging, such as the Nokia Lumia 920, Samsung Galaxy S4, Google Nexus 4 and HTC Droid DNA
- Wireless charging sleeves for Galaxy S3, iPhone 4S, from brands like Energizer, Phillips and Duracell
- Charging pads for mobile phones, from makers like Powermat, Samsung, LG, JBL and Nokia
- Devices that double up as charging pads, such as JBL PowerUp, Oregon Scientific Bedside Alarm
- Wireless charging video cameras, e.g., the Qi-enabled Panasonic HC-V720M, Pentax WG-3 GPS
- Low-power consumer goods like electric toothbrushes (e.g., Oral-B rechargeable toothbrushes)
- Smart watches, such as the new Agent by Secret Labs
- Electric charging in vehicles like the Nissan Leaf, Chevrolet Volt and Renault Fluence using technologies like Plugless L2, Qualcomm Halo, etc
- Wireless charging kitchen counters from DuPont
- Kitchen appliances that can be wirelessly charged, such as blender and rice cooker from Haier
- Intel proposes A4WP-based laptops, tablets, etc, in the near future
- Navigation devices, such as those demonstrated by Fulton Innovation at CES 2013

of hygiene and ergonomics, completely-sealed devices are preferred.

For the benefit of the consumer electronics industry, the A4WP specification goes with broadly-adopted wireless technologies, such as Bluetooth LE, which will allow manufacturers to minimise hardware requirements. For industrial designers, the A4WP specification leverages a near-field magnetic resonance technology that provides more flexibility for charging applications to be installed into automobiles, furniture and other surfaces.

The A4WP specification also features a simpler transmitter antenna design, a simpler wireless power control system, and the ability to transfer power through non-metallic surfaces at greater distances, without the need for costly multi-coil repeaters. Besides, A4WP promises that the specification will only freeze the requirements for the transmitter and receiver, mutual coupling and mutual inductance, leaving most options open to members' designers. Designers can specify and source their own out-of-

band radios, power amplifiers, DC-to-DC converters, rectifiers and microprocessors—discrete or integrated—and assemble them as they require. As long as the components conform to the specification, they can be of any topology.

It is believed that A4WP's specification is based on WiPower, Qualcomm's wireless charging specification, which was approved by the A4WP in January. A4WP is supported by other industry leaders including Intel, IDT, TI, Broadcom, Samsung, LG, Haier, Denso, and so on.

Charge anywhere in a room with WiTricity

The PMA recently formed a technical working group to add a resonant implementation to the Power 2.0 group of specifications. WiTricity, an MIT spin-off and the exclusive patentee for MIT's wireless energy transfer technology, was appointed vice-chair of this working group, with the immediate task of creating a specification for smartphone-ready highly-resonant wireless power.

This specification, expected to be available by the end of this year, will be fully compatible with the PMA's existing inductive wireless power specification and products.

WiTricity's wireless charging technology tops the benefits

of spatial freedom, by offering distanced wireless charging using tuned electromagnetic resonators that efficiently transfer power over large distances via the magnetic near field. So, you just need to walk into a WiTricity powered location, and your devices will start charging. Although it might seem that WiTricity's technology is similar to traditional magnetic induction, it is not. In traditional systems, the efficiency of the power exchange drops by orders of magnitude when the distance between the coils becomes larger than their sizes. As we have seen earlier in this feature, the power exchange efficiency of some induction systems is improved using resonant circuits. However, the technical literature on WiTricity's website claims that, "...to the best of our knowledge, WiTricity's founding technical team was the first to discover that by specially designing the magnetic resonators, one could achieve strong coupling and highly-efficient energy exchange over distances much larger than the size of the resonator coils, distances very large compared to traditional schemes."

WiTricity's mode of wireless power transfer can work over distances ranging from centimetres to several metres—and that too with a very high efficiency, often exceeding 90 per cent. Another interesting aspect is that WiTricity sources only transfer energy when it is needed. When a WiTricity powered device no longer needs to capture additional energy, the WiTricity power source will automatically reduce its power consumption to a power-saving state.

WiTricity's method of energy transfer can penetrate and wrap around obstacles, as it uses magnetic near field. It is also a zero-radiation, safe method of energy transfer. WiTricity also features a scalable design that enables solutions from milli-watts to kilo-watts.

Putting the technology into your devices

So, how does all this technology translate into products? Well, we already have wireless charging phones and dozens of mats and pads. There are also wireless charging sleeves for various



Nokia announces JBL PowerUp wireless charging speaker for Nokia Lumia